Lecture 6: Direct Preference Optimization

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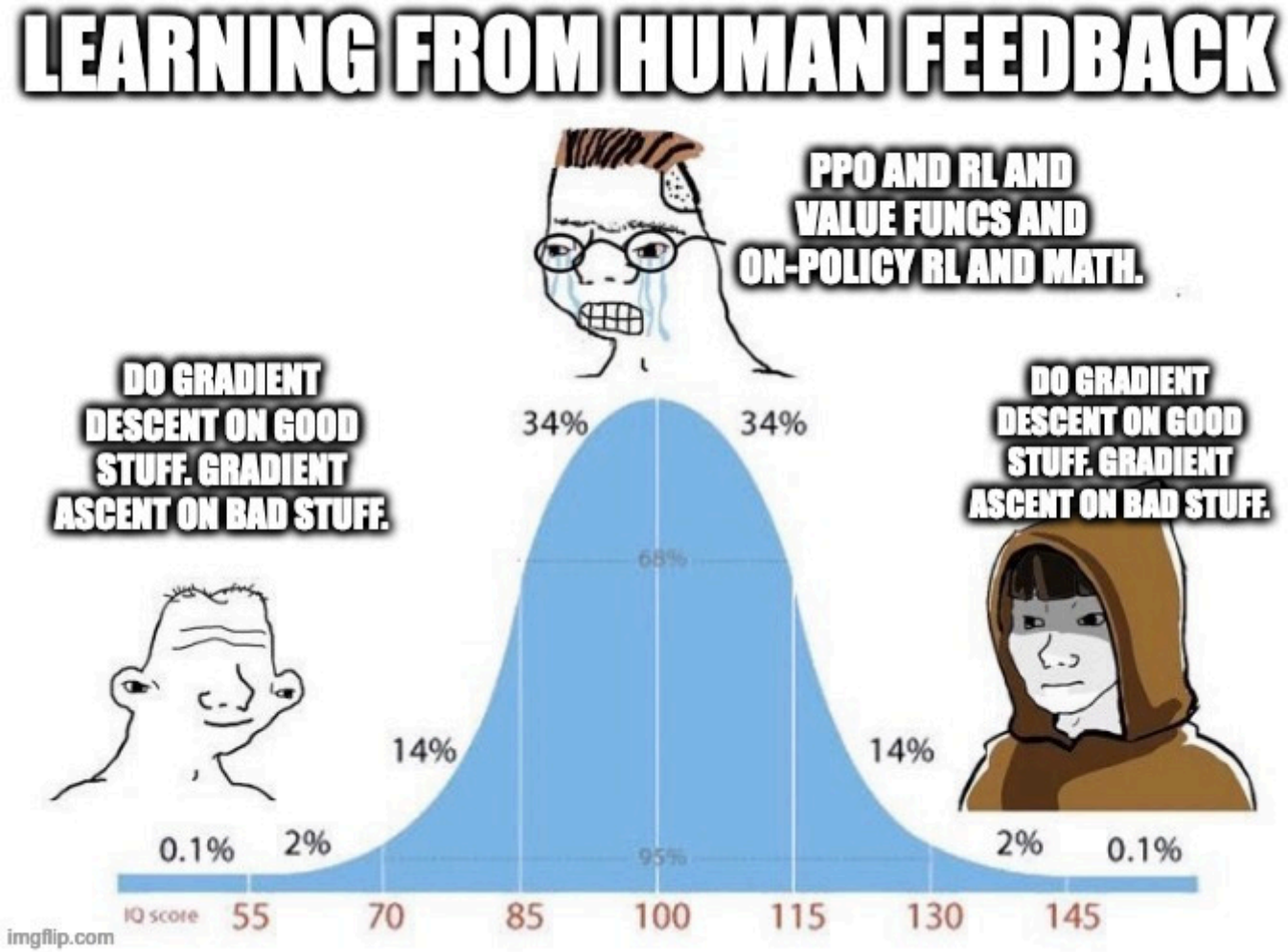
Post-training course. Chapter 8 on Direct Alignment Algorithms.

Why DPO was – and is – such a big deal

The Direct Preference Optimization paper (Rafailov et al., 2023) was a breakthrough in the accessibility of RLHF and post-training research. Some context:

- DPO paper was released in May of 2023, when many groups were struggling to get open-source replications of RLHF pipelines going. Most "aligned" models were just SFT.
- It took until about the fall of 2023 for people to figure out the right data and settings for DPO to work
- Zephyr-Beta, Tülu 2, and other models that fall opened the floodgates of post-training research. Iterating on DPO methods in 2024 felt as exciting as RL methods in 2025 or tool-use methods in 2026.
- **Terminology note:** I call the class of algorithms that operate like DPO on static preference data Direct Alignment Algorithms (DAAs). The name didn't stick much, but we'll see there are a ton of variants!

The DPO debate (DPO v RL)



When DPO was released it sparked a fierce debate about how best to do RLHF and preference learning. Meme credit: Tom Goldstein.

This lecture

We will **derive Direct Preference Optimization (DPO) from scratch**, then look at how it is used in practice.

The promise of DPO:

- No separate reward model
- No reinforcement learning loop
- Just a single, directly-differentiable loss on preference pairs

The plan, in four steps:

1. Solve the RLHF objective for its **optimal policy**
2. Invert that to write the **reward in terms of the policy**
3. Plug it into the **Bradley-Terry** preference model
4. Read off the **DPO loss** and its **gradient**

Rafailov et al., 2023 — "Your Language Model is Secretly a Reward Model" (Rafailov et al., 2023).

Recall: Where DPO sits in the pipeline

Classic RLHF is three moving parts:

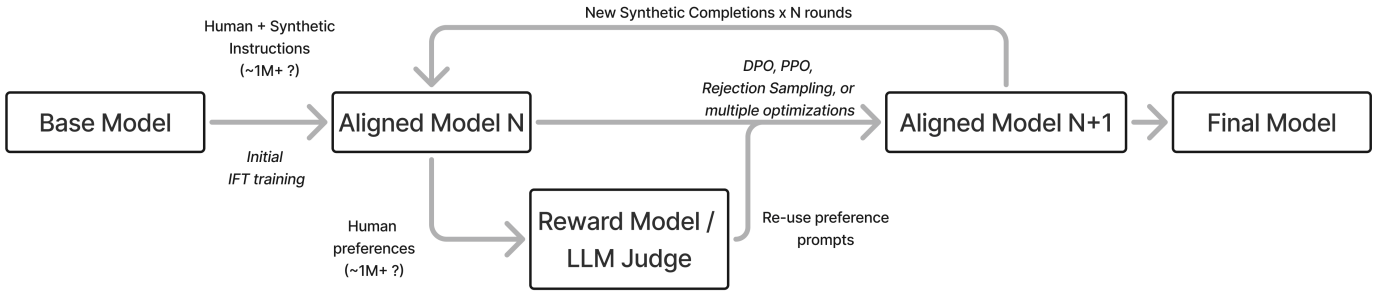
1. Collect human preference pairs
2. Train a reward model $r_\phi(x, y)$
3. Optimize the policy against r_ϕ with RL (e.g. PPO), under a KL penalty

DPO collapses steps 2 and 3 into **one supervised-style loss**.

The key realization we will prove:

> *The optimal RLHF policy and the reward model are two views of the same object. If we know one, we know the other in closed form.*

So we can train the policy *directly* on preferences.



Modern post-training runs many rounds of feedback and optimization; DPO is one of the interchangeable optimizers in that loop.

DPO at a glance: Where we're headed

The whole derivation lands on two objects.

Optimal policy — the RLHF solution in closed form:

$$\pi^*(y \mid x) = \frac{1}{Z(x)}\,\pi_{\text{ref}}(y \mid x)\exp\!\left(\tfrac{1}{\beta} r(x, y)\right)$$

DPO loss — what you actually train:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)} - \beta \log \frac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)}\right)\right]$$

In practice, that loss is just a few lines:

```
pi_logratios  = policy_chosen_logps  -
policy_rejected_logps
ref_logratios = reference_chosen_logps -
reference_rejected_logps
logits = pi_logratios - ref_logratios
losses = -F.logsigmoid(beta * logits)
```

Everything between here and the implementation is *why* this is the right loss.

DPO derivation P1/4: Deriving the optimal policy

Start with the RLHF optimization problem

$$\max_{\pi} \; \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi}\Big[\, r(x,y) \,\Big] - \beta\, \mathcal{D}_{\mathrm{KL}}\Big(\pi \,\|\, \pi_{\mathrm{ref}}\Big) \quad \text{the RLHF objective}$$

We want to find the π that solves this equation! But, without RL.

Fold the KL into the expectation

$$\max_{\pi} \; \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi}\Big[\, r(x,y) \,\Big] - \beta\, \mathcal{D}_{\mathrm{KL}}\big(\pi \,\|\, \pi_{\mathrm{ref}}\big) \qquad \text{the RLHF objective}$$

$$= \max_{\pi} \; \mathbb{E}\Big[\, r(x,y) - \beta \log \tfrac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)} \,\Big] \qquad \text{write KL} = \mathbb{E}_{x\sim\mathcal{D},y\sim\pi}\Big[\log \tfrac{\pi}{\pi_{\mathrm{ref}}}\Big]$$

Fold the KL into the expectation

$$\max_{\pi} \; \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi}\big[\, r(x,y)\,\big] - \beta\, \mathcal{D}_{\mathrm{KL}}\big(\pi \,\|\, \pi_{\mathrm{ref}}\big) \qquad \text{the RLHF objective}$$

$$= \max_{\pi} \; \mathbb{E}\Big[\, r(x,y) - \beta \log \tfrac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)} \,\Big] \qquad \text{write KL} = \mathbb{E}_{x\sim\mathcal{D},y\sim\pi}\big[\log \tfrac{\pi}{\pi_{\mathrm{ref}}}\big]$$

$$= \max_{\pi} \Big(\mathbb{E}[r(x,y)] - \beta\, \mathbb{E}\big[\log \tfrac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)}\big] \Big) \qquad \text{split into two terms}$$

Flip to a minimization & clean-up

$$\max_{\pi} \left(\mathbb{E}[r(x,y)] - \beta \, \mathbb{E}\Big[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \Big] \right) \quad \text{where we left off}$$

Flip to a minimization & clean-up

$$\max_{\pi} \left(\mathbb{E}[r(x,y)] - \beta \, \mathbb{E}\Big[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \Big] \right) \qquad \text{where we left off}$$

$$= \min_{\pi} \left(-\mathbb{E}[r(x,y)] + \beta \, \mathbb{E}\Big[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \Big] \right) \qquad \text{multiply by } -1 : \ \max \to \min$$

Flip to a minimization & clean-up

$$\max_{\pi} \left(\mathbb{E}[r(x,y)] - \beta \, \mathbb{E}\Big[\log \tfrac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \Big] \right) \qquad \text{where we left off}$$

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$$= \min_{\pi} \mathbb{E}\Big[\log \tfrac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \tfrac{1}{\beta} r(x,y) \Big] \qquad \text{divide by } \beta, \text{ recombine}$$

Fold the optimization target into one log-ratio

Pick up where we left off and fold the bracket into a single log-ratio, using $\frac{1}{\beta} r = \log e^{r/\beta}$:

$$\min_{\pi} \mathbb{E}\Big[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x, y) \Big] \quad \text{where we left off}$$

$$= \min_{\pi} \mathbb{E}\Big[\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)\, e^{\, r(x,y)/\beta}} \Big]$$

- π is now measured against $\pi_{\text{ref}}\, e^{r/\beta}$: the **reference reweighted by reward** — each response's probability is multiplied by $e^{r/\beta}$, which is larger the higher its reward (β sets how aggressive the reweighting is).
- If that target were a probability distribution q, this $\min_\pi$ would be a **KL divergence**, $\mathbb{E}_{y\sim\pi}\big[\log \frac{\pi}{q}\big] = \mathcal{D}_{\text{KL}}(\pi \,\|\, q)$ — minimized *exactly*, in closed form (Gibbs' inequality).

Introduce the partition function $Z(x)$

We are minimizing $\min_\pi \; \mathbb{E}_{y\sim\pi}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)\, e^{\, r(x,y)/\beta}} \right].$

Why do we want π to match that denominator?

Introduce the partition function $Z(x)$

We are minimizing $\min_\pi \; \mathbb{E}_{y\sim\pi}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)\, e^{\, r(x,y)/\beta}} \right]$.

Why do we want π to match that denominator? Because $\mathbb{E}_{y\sim\pi}\left[\log \frac{\pi}{q}\right]$ is a **KL divergence** $\mathcal{D}_{\text{KL}}(\pi \,\|\, q)$ — it is ≥ 0, and equals 0 **only when** $\pi = q$. So the minimizer is whatever distribution q sits in the denominator.

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The catch: our denominator $\pi_{\text{ref}}\, e^{r/\beta}$ is **not a distribution** — summed over y it does not equal 1.

Introduce the partition function $Z(x)$

We are minimizing $\min_\pi \; \mathbb{E}_{y\sim\pi}\left[\, \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)\, e^{\,r(x,y)/\beta}} \,\right]$.

Why do we want π to match that denominator? Because $\mathbb{E}_{y\sim\pi}\left[\log \frac{\pi}{q}\right]$ is a **KL divergence** $\mathcal{D}_{\text{KL}}(\pi \,\|\, q)$ — it is ≥ 0, and equals 0 **only when** $\pi = q$. So the minimizer is whatever distribution q sits in the denominator.

The catch: our denominator $\pi_{\text{ref}}\, e^{r/\beta}$ is **not a distribution** — summed over y it does not equal 1. We normalize it with the partition function:

$$Z(x) = \sum_y \pi_{\text{ref}}(y \mid x)\, \exp\!\left(\tfrac{1}{\beta} r(x,y)\right) \qquad (\text{depends on } x \text{ and } r, \text{ not on } \pi)$$

- Now $q(y \mid x) = \frac{1}{Z(x)}\, \pi_{\text{ref}}(y \mid x)\, e^{\,r/\beta}$ **is** a valid distribution.
- The objective becomes a genuine $\mathcal{D}_{\text{KL}}(\pi \,\|\, q)$ — minimized exactly at $\pi = q$ (made rigorous with Gibbs' inequality shortly).
- Next we massage the inside of the objective so $Z(x)$ appears.

Revisiting the objective with the partition function $Z(x)$

$$\min_{\pi} \mathbb{E}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x,y)\right] \quad \text{where we left off}$$

Revisiting the objective with the partition function $Z(x)$

$$\min_{\pi} \mathbb{E}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x,y)\right] \qquad \text{where we left off}$$

$$= \min_{\pi} \mathbb{E}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x,y) + \log Z(x) - \log Z(x)\right] \qquad \text{add } 0 = \log Z - \log Z$$

Revisiting the objective with the partition function $Z(x)$

$$\min_{\pi} \mathbb{E}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x,y)\right] \qquad \text{where we left off}$$

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$$= \min_{\pi} \mathbb{E}\left[\left(\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} + \log Z(x)\right) - \log Z(x) - \frac{1}{\beta} r(x,y)\right] \qquad \text{group terms}$$

Revisiting the objective with the partition function $Z(x)$

$$\min_{\pi} \mathbb{E}\left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x,y)\right] \qquad \text{where we left off}$$

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$$= \min_{\pi} \mathbb{E}\left[\left(\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} + \log Z(x)\right) - \log Z(x) - \frac{1}{\beta} r(x,y)\right] \qquad \text{group terms}$$

$$= \min_{\pi} \mathbb{E}\left[\log \frac{\pi(y \mid x)}{\frac{1}{Z(x)}\pi_{\text{ref}}(y \mid x)} - \log Z(x) - \frac{1}{\beta} r(x,y)\right] \qquad \log a + \log b = \log ab$$

Turn the objective into a KL divergence

Fold $\frac{1}{\beta} r = \log e^{r/\beta}$ into the denominator of the first term in the objective (again, log rules) and name it $q(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}$ — a valid distribution over y. The objective is:

$$\min_{\pi} \; \mathbb{E}_{x \sim \mathcal{D}} \, \mathbb{E}_{y \sim \pi} \Big[\log \tfrac{\pi(y|x)}{q(y|x)} - \log Z(x) \Big]$$

$\log Z(x)$ does not depend on y, so pull it out of the inner expectation:

$$\min_{\pi} \; \mathbb{E}_{x \sim \mathcal{D}} \Big[\mathbb{E}_{y \sim \pi} \big[\log \tfrac{\pi(y|x)}{q(y|x)} \big] - \log Z(x) \Big]$$

The inner expectation is exactly a **KL divergence**, $\mathbb{E}_{y \sim \pi} \big[\log \frac{\pi}{q} \big] = \mathcal{D}_{\text{KL}}(\pi \,\|\, q)$:

$$\min_{\pi} \; \mathbb{E}_{x \sim \mathcal{D}} \big[\mathcal{D}_{\text{KL}}\big(\pi(y \mid x) \,\|\, q(y \mid x)\big) - \log Z(x) \big]$$

Apply Gibbs' inequality to read off π^*

Where we landed – the objective is now a KL plus a π-independent constant:

$$\min_{\pi} \; \mathbb{E}_{x\sim\mathcal{D}}\Big[\, \mathcal{D}_{\mathrm{KL}}\big(\pi(y \mid x) \,\|\, q(y \mid x)\big) - \log Z(x) \,\Big]$$

- $\log Z(x)$ does not depend on π — only the KL term can be optimized.
- A KL is ≥ 0 and equals 0 **only when** $\pi = q$ (Gibbs' inequality), so the minimizer is $\pi^* = q$:

$$\boxed{\pi^*(y \mid x) = q(y \mid x) = \tfrac{1}{Z(x)}\, \pi_{\mathrm{ref}}(y \mid x)\, \exp\big(\tfrac{1}{\beta} r(x, y)\big)}$$

DPO derivation P2/4: Recovering the reward from the policy (needed to implement it)

Invert π^* for the implicit reward

$$\pi^*(y \mid x) = \frac{1}{Z(x)}\, \pi_{\text{ref}}(y \mid x)\, \exp\Big(\frac{1}{\beta} r^*(x, y)\Big)$$

Invert π^* for the implicit reward

$$\log \pi^*(y \mid x) = \log\left(\frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r^*(x, y)\right)\right) \quad \text{take log of both sides}$$

Invert π^* for the implicit reward

$$\log \pi^*(y \mid x) = \log\left(\tfrac{1}{Z(x)} \, \pi_{\text{ref}}(y \mid x) \, \exp\left(\tfrac{1}{\beta} r^*(x, y)\right)\right) \qquad \text{take } \log \text{ of both sides}$$

$$\log \pi^*(y \mid x) = -\log Z(x) + \log \pi_{\text{ref}}(y \mid x) + \tfrac{1}{\beta} r^*(x, y) \qquad \log abc = \log a + \log b + \log c$$

Invert π^* for the implicit reward

$$\log \pi^*(y \mid x) = \log\left(\tfrac{1}{Z(x)} \, \pi_{\text{ref}}(y \mid x) \, \exp\left(\tfrac{1}{\beta} r^*(x, y) \right) \right) \qquad \text{take } \log \text{ of both sides}$$

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$$\tfrac{1}{\beta} r^*(x, y) = \log \pi^*(y \mid x) - \log \pi_{\text{ref}}(y \mid x) + \log Z(x) \qquad \text{rearrange for } r^*$$

Invert π^* for the implicit reward

$$\log \pi^*(y \mid x) = \log\left(\tfrac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\tfrac{1}{\beta} r^*(x, y)\right)\right) \qquad \text{take } \log \text{ of both sides}$$

$$\log \pi^*(y \mid x) = -\log Z(x) + \log \pi_{\text{ref}}(y \mid x) + \tfrac{1}{\beta} r^*(x, y) \qquad \log abc = \log a + \log b + \log c$$

$$\tfrac{1}{\beta} r^*(x, y) = \log \pi^*(y \mid x) - \log \pi_{\text{ref}}(y \mid x) + \log Z(x) \qquad \text{rearrange for } r^*$$

$$r^*(x, y) = \beta \log \tfrac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x) \qquad \text{multiply by } \beta$$

$$\boxed{r^*(x, y) = \beta \log \tfrac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)}$$

your LM is secretly a reward model

DPO derivation P3/4: Connecting to Bradley-Terry preference (what our alignment data has looked like)

Recall the Bradley-Terry model

A preference between two responses is a softmax over their rewards:

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp\big(r^*(x, y_1)\big)}{\exp\big(r^*(x, y_1)\big) + \exp\big(r^*(x, y_2)\big)}$$

Now substitute the implicit reward $r^*(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$.

New here? See Lecture 2 and Chapter 5: Reward Modeling for the Bradley-Terry model.

Substitute the reward, then cancel $Z(x)$

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} + \beta \log Z(x)\right)}{\exp\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} + \beta \log Z(x)\right) + \exp\left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} + \beta \log Z(x)\right)}$$

- Every term carries the same factor $\exp(\beta \log Z(x)) = Z(x)^\beta$.

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}\right)}{\exp\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}\right) + \exp\left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)}\right)}$$

- Decompose $e^{a+b} = e^a e^b$ and cancel the shared $Z(x)^\beta$ — the **intractable term is gone**.

Divide through to a sigmoid

Write $\Delta_i = \beta \log \frac{\pi^*(y_i|x)}{\pi_{\text{ref}}(y_i|x)}$, so the cancelled form is $p^* = \frac{e^{\Delta_1}}{e^{\Delta_1} + e^{\Delta_2}}$. Multiply numerator and denominator by $e^{-\Delta_1}$:

$$p^*(y_1 \succ y_2 \mid x) = \frac{e^{\Delta_1}\, e^{-\Delta_1}}{e^{\Delta_1}\, e^{-\Delta_1} + e^{\Delta_2}\, e^{-\Delta_1}} = \frac{1}{1 + e^{\,\Delta_2 - \Delta_1}}$$

- The numerator becomes 1; the denominator holds an exponential of the **difference** $\Delta_2 - \Delta_1$.

With $\sigma(z) = \frac{1}{1+e^{-z}}$, this is a sigmoid ($\Delta_i = \beta \log \frac{\pi^*(y_i|x)}{\pi_{\text{ref}}(y_i|x)}$):

$$\boxed{p^*(y_1 \succ y_2 \mid x) = \sigma\left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} - \beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)}\right)}$$

- A **sigmoid of the difference of two log-ratios** — Bradley-Terry, with the policy in place of the reward model.

DPO derivation P4/4: The loss function and gradient

The DPO loss

Minimize the negative log-likelihood of the observed preferences $(y_c \succ y_r)$ – this is making the probability more likely:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_c,y_r)\sim\mathcal{D}}\left[\log p(y_c \succ y_r \mid x)\right]$$

Plug in the sigmoid form of p, with the trainable policy π_θ as the implicit reward:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,y_c,y_r)\sim\mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)} - \beta \log \frac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)}\right)\right]$$

This is directly differentiable — **no reward model, no sampling, no RL loop.**

The DPO gradient

Start by taking the gradient of the previous loss function.

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\nabla_\theta \, \mathbb{E}_{(x,y_c,y_r)}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)} - \beta \log \frac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)}\right)\right]$$

Let $u = \left(\beta \log \frac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)} - \beta \log \frac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)}\right)$ — the term inside σ.

Chain rule, starting with $\frac{d}{dz}\log\sigma(z) = \frac{\sigma'(z)}{\sigma(z)}$ as the inner operation:

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,y_c,y_r)}\left[\frac{\sigma'(u)}{\sigma(u)} \nabla_\theta u\right]$$

Substitute $\sigma'(u) = \sigma(u)\big(1-\sigma(u)\big)$ and cancel $\sigma(u)$:

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,y_c,y_r)}\left[\frac{\sigma(u)\big(1-\sigma(u)\big)}{\sigma(u)} \nabla_\theta u\right] = -\mathbb{E}_{(x,y_c,y_r)}\left[\big(1-\sigma(u)\big) \nabla_\theta u\right]$$

The DPO gradient

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,y_c,y_r)}\Big[\big(1-\sigma(u)\big)\,\nabla_\theta u\Big]$$

The reflection identity $1-\sigma(u)=\sigma(-u)$ flips the argument, giving the compact form:

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x,y_c,y_r)}\Big[\sigma(-u)\,\nabla_\theta u\Big]$$

Now differentiate u, where $-u = \beta \log \frac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)} - \beta \log \frac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)}$. Only π_θ depends on θ — π_{ref} is frozen, so its $\log$ terms drop:

$$\nabla_\theta u = \beta\big[\nabla_\theta \log \pi_\theta(y_c \mid x) - \nabla_\theta \log \pi_\theta(y_r \mid x)\big]$$

Substitute $\nabla_\theta u$ and write $\sigma(-u)$ out in full:

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\beta\,\mathbb{E}_{(x,y_c,y_r)}\Big[\sigma\big(\beta \log \tfrac{\pi_\theta(y_r|x)}{\pi_{\text{ref}}(y_r|x)} - \beta \log \tfrac{\pi_\theta(y_c|x)}{\pi_{\text{ref}}(y_c|x)}\big)\,\big[\nabla_\theta \log \pi_\theta(y_c \mid x) - \nabla_\theta \log \pi_\theta(y_r \mid x)\big]\Big]$$

The DPO gradient

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\beta\, \mathbb{E}_{(x,y_c,y_r)}\Big[\underbrace{w}_{\text{how wrong}} \cdot \big(\nabla_\theta \log \pi_\theta(y_c \mid x) - \nabla_\theta \log \pi_\theta(y_r \mid x)\big) \Big]$$

$$\text{where} \quad w = \sigma\big(r_\theta(x, y_r) - r_\theta(x, y_c)\big)$$

- **Weight** $w \in (0, 1)$ is larger when the model is *more wrong* — when it ranks the rejected response above the chosen the loss is higher.
- **The delta-gradient bracket** raises the likelihood of y_c and lowers that of y_r.
- β scales the step, trading correct ordering against drift from π_{ref} (downstream of the KL penalty in the RLHF objective).

Recap: The DPO derivation

$$\textbf{Objective} \;:\; \max_{\pi} \mathbb{E}\big[r(x,y)\big] - \beta\, \mathcal{D}_{\mathrm{KL}}(\pi \,\|\, \pi_{\mathrm{ref}})$$

$$\Rightarrow\; \textbf{Optimal policy} \;:\; \pi^*(y \mid x) = \tfrac{1}{Z(x)}\, \pi_{\mathrm{ref}}(y \mid x) \exp\big(\tfrac{1}{\beta} r(x,y)\big)$$

$$\Rightarrow\; \textbf{Implicit reward} \;:\; r^*(x,y) = \beta \log \tfrac{\pi^*(y|x)}{\pi_{\mathrm{ref}}(y|x)} + \beta \log Z(x)$$

$$\Rightarrow\; \textbf{Bradley-Terry} \;:\; p^*(y_1 \succ y_2) = \sigma\big(r^*(x,y_1) - r^*(x,y_2)\big) \;\; (Z \text{ cancels})$$

$$\Rightarrow\; \textbf{DPO loss} \;:\; -\log \sigma\big(\beta \log \tfrac{\pi_\theta(y_c)}{\pi_{\mathrm{ref}}(y_c)} - \beta \log \tfrac{\pi_\theta(y_r)}{\pi_{\mathrm{ref}}(y_r)}\big)$$

The reward model never had to be built — it was hiding inside the policy the whole time.

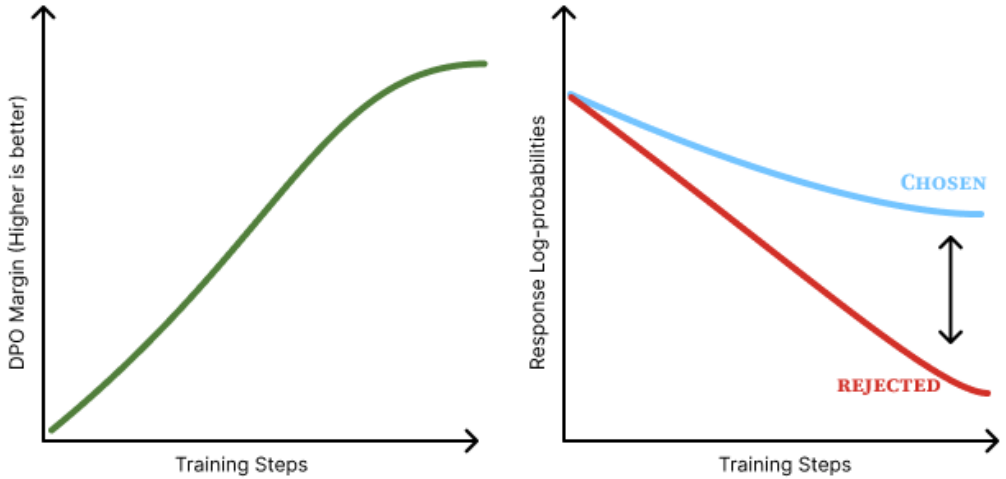
DPO weaknesses, variants, and implementation details

A subtle risk: The chosen probability can fall

The DPO loss only cares about the **margin** between the chosen and rejected log-ratios — not their absolute values. So the model can lower the loss by pushing the *rejected* probability down **faster** than the chosen, even while the **chosen probability also falls**.

Mediated through the partition function $Z(x)$ in the derivation.

- Called **likelihood displacement** (Razin et al., 2025) (Ren & Sutherland, 2025); posited to push probability toward unaddressed, off-distribution behaviors.
- A reason some practitioners add an SFT term on the chosen response, or use fixes like Cal-DPO (Xiao et al., 2024) / AlphaPO (Gupta et al., 2025).



Sketch of likelihood displacement in DPO.

In a real run: Olmo 1B DPO — the chosen/rejected reward margin widens, yet the *chosen* log-prob itself can still drift down.

The β parameter and the static KL

β sets the strength of the KL constraint relative to reward maximization:

- **Large** $\beta \rightarrow$ policy stays close to π_{ref}; it barely moves.
- **Small** $\beta \rightarrow$ policy is free to deviate; it can **over-optimize**.

Crucially, DPO's final KL distance is **static**: it steps directly to the *optimal* solution implied by the dataset and the chosen β. Online RL instead takes steps based on freshly sampled batches and a per-sample KL penalty. Some RL runs even include dynamically adjusted KL controllers.

A zoo of direct alignment algorithms

Each variant tweaks the loss to fix a limitation — often a one-line change. I started calling all the variants Direct Alignment Algorithms (DAAs). Two to start:

- **Identity Preference Optimization (IPO)** (Azar et al., 2024) — softens the preference probability away from Bradley-Terry to curb overfitting.

$$\mathcal{L}_{\text{IPO}} = \mathbb{E}_{(x,y_w,y_l)}\left[\left(\log \frac{\pi_\theta(y_w|x)\,\pi_{\text{ref}}(y_l|x)}{\pi_\theta(y_l|x)\,\pi_{\text{ref}}(y_w|x)} - \frac{1}{2\beta}\right)^2\right]$$

- **Conservative DPO (cDPO)** (Rafailov et al., 2023) (in the original DPO paper) — assumes the preference labels are noisy (flipped with probability ε) and softens the loss target accordingly.
- **Offset DPO (ODPO)** (Amini et al., 2024) — requires the chosen to beat the rejected by a margin offset, scaled by how strong the preference is.

A zoo of direct alignment algorithms

Two more that **drop the reference model** entirely:

- **Odds Ratio Preference Optimization (ORPO)** (Hong et al., 2024) — adds an odds-ratio pull toward the chosen response, folded directly into the SFT loss. The odds ratio uses only π_θ, which allows dropping the reference model.

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} - \lambda \mathbb{E}\left[\log \sigma\left(\log \frac{\text{odds}_\theta(y_w|x)}{\text{odds}_\theta(y_l|x)}\right)\right], \quad \text{odds}_\theta(y) = \frac{\pi_\theta(y)}{1-\pi_\theta(y)}$$

- **Simple Preference Optimization (SimPO)** (Meng et al., 2025) — length-normalizes the reward into an average log-prob, with a target margin γ. $\frac{1}{|y|}$ normalizes by length; γ is a target margin; no π_{ref}.

$$\mathcal{L}_{\text{SimPO}} = -\mathbb{E}\left[\log \sigma\left(\frac{\beta}{|y_w|}\log \pi_\theta(y_w \mid x) - \frac{\beta}{|y_l|}\log \pi_\theta(y_l \mid x) - \gamma\right)\right]$$

The algorithm matters **far less** than the base model and the data. Still, many papers continued to make minor algorithmic tweaks. Many more exist than were on these slides.

Implementation is genuinely simple

No generation during training, no separate reward model — the heart of the loss is a few lines:

```
# log-prob gaps for policy and frozen reference
pi_logratios  = policy_chosen_logps - policy_rejected_logps
ref_logratios = reference_chosen_logps - reference_rejected_logps

# positive when policy shifts mass toward the chosen completion
logits = pi_logratios - ref_logratios
losses = -F.logsigmoid(beta * logits)
```

Tip: π_{ref} is frozen, so precompute and cache its log-probs to cut peak memory ~50%. Reference code: `code/direct_alignment/`.

DAAs work with synthetic preference data

These algorithms need *feedback* data, not necessarily *human* feedback data — AI feedback works just as well.

- Most modern DPO uses preferences labeled by a strong model. **UltraFeedback** (Cui et al., 2023) was the first prominent *open* synthetic preference dataset and helped kickstart the DPO era; Tülu 3 (Lambert et al., 2024), SmolLM 2 (Ben Allal et al., 2025), and others followed with larger synthetic-feedback recipes.
- **On-policy data** (some completions from the model you are tuning) helps the contrastive loss optimize the right token space within a complex post-training recipe (studied this specifically in Tülu 3).
- **Delta Learning** (Geng et al., 2025): later work argues the *gap* between chosen and rejected matters more than which models produced them (e.g. Qwen3-32B chosen vs Qwen3-0.6B rejected). Used this in Olmo 3!

DPO vs. RL: Offline vs. online

DPO and other DAAs

- Train on a fixed dataset collected ahead of time.
- Simpler, more stable, fast to iterate on data.
- Limited by the **coverage** of that dataset — a slightly lower performance ceiling.

PPO / policy gradient is online

- Generate fresh completions during training, score with a reward model.
- Can explore new regions → often higher peak performance (Ivison et al., 2024) (Xu et al., 2024) (Tajwar et al., 2024).
- More compute, more moving parts (four models in memory), more engineering complexity.

Where does this leave us today?

If so many models have used DPO well and it's so simple, why does it seem like it comes up so infrequently – especially at the frontier?

Hypotheses for DPO's role today

- **Olmo 3** (Nov 2025) / **SmolLM 3** (Jul 2025) – used DPO over reasoning traces after SFT to boost performance in a simple pipeline.
- **NVIDIA Nemotron 3** (Dec 2025) – *Mixed Preference Optimization*: DPO + Binary Classifier Optimization in one offline stage, over data scored by a generative reward model.
- **Liquid AI LFM2** (Nov 2025) – length-normalized DPO on semi-online data, before a final RLVR pass.

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How I see things:

- DPO works well in wonky, distillation-heavy recipes like Olmo – e.g. a model with a spikier distribution, with training data from many teacher models.
- DPO still works in other settings, but most labs have the engineering resources to do other things with higher peak performance.
- TLDR: DPO is a path to a good/solid model, but not to the best model. And DPO may do less in *cleaner* training recipes.

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